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Wearable image display device

Abstract

A wearable image display device includes a display unit that displays an image and a wearing unit attached to the display unit. The display unit includes a first positioning portion and a first rotation mechanism that rotates relative to the wearing unit in a rotational direction with a first direction serving as a rotation axis. The wearing unit includes a second positioning portion brought into contact with the first positioning portion and a first fitting mechanism fitted to the first rotation mechanism. The first positioning portion and the second positioning portion are brought into contact with each other either at a first position or at a second position different from the first position in the rotational direction.

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2021-127767, filed Aug. 3, 2021, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a wearable image display device that presents a virtual image to a wearer. In particular, the present disclosure relates to a wearable image display device capable of adjusting a wearing position or a display position of the virtual image.

2. Related Art

(3) As a wearable image display device, one that includes a base unit having a pair of temple members and a movable unit having a pair of display units is known, wherein the base unit includes a pair of hinges that rotatably support the temple members, and the movable unit is rotatably journaled by the base unit with a pair of movable portions attached in the vicinity of the

pair of hinges serving as fulcrums, and can be raised and lowered in front of the eyes (JP-A-2013-214910).

(4) The device of JP-A-2013-214910 has a structure in which the operation portion provided at the center of the movable unit is bent to reduce frictional resistance of the sliding surface and thereby facilitate vertical position adjustment. Thus, even when the movable unit is stable at the lower end position, the movable unit may not be stable at upper positions, so the display unit may be displaced from the position suitable for the wearer's eyes.

SUMMARY

(5) A wearable image display device according to one aspect of the present disclosure includes: a display unit that displays an image, and a wearing unit attached to the display unit, wherein the display unit includes a first positioning portion and a first rotation mechanism that rotates relative to the wearing unit in a rotational direction with a first direction serving as a rotation axis, the wearing unit includes a second positioning portion brought into contact with the first positioning portion and a first fitting mechanism fitted to the first rotation mechanism, and the first positioning portion and the second positioning portion are brought into contact with each other either at a first position or at a second position different from the first position in the rotational direction.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view illustrating a wearable image display device according to a first embodiment.

(2) FIG. 2 is a perspective view illustrating a wearable image display device.

(3) FIG. 3 is a side view illustrating a wearing state of a wearable image display device.

(4) FIG. 4 is a partially cutaway perspective view of a first temple.

(5) FIG. 5 is an exploded perspective view of the first temple.

(6) FIG. 6 is a partially cutaway perspective view of a second temple.

(7) FIG. 7 is an exploded perspective view of the second temple.

(8) FIG. 8 is a plan view illustrating an optical device of a wearable image display device.

(9) FIG. 9 is a plan view illustrating an optical structure of a first image-forming device.

(10) FIG. 10 is a side view illustrating adjustment of the disposition of a temple.

(11) FIG. 11 is a side view illustrating adjustment of the disposition of a temple.

(12) FIG. 12 is a side view illustrating adjustment of the display direction by a display device.

(13) FIG. 13 is a plan view and the like illustrating the appearance of a wearable image display device according to a second embodiment.

(14) FIG. 14 is a partially enlarged perspective view of a wearable image display device.

(15) FIG. 15 is a view illustrating a cross section of a wearable image display device.

(16) FIG. 16 is a partially cutaway cross-sectional view illustrating an angle adjustment mechanism.

(17) FIG. 17 is a side cross-sectional view illustrating an optical structure of the first image-forming device.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

First Embodiment

(18) Hereinafter, a first embodiment of a wearable image display device according to the present disclosure will be described with reference to drawings.

(19) Referring to FIGS. 1 to 3, a head-mounted display (hereinafter also referred to as “HMD”) 100 is a wearable image display device, and allows an observer wearing the same or a wearer US to recognize video as a virtual image. In FIG. 1 and the like, X, Y, and Z represent an orthogonal coordinate system: the +X direction corresponds to a lateral direction in which the eyes EY of the

wearer US wearing the HMD **100** are aligned; the +Y direction corresponds to an upward direction orthogonal to the lateral direction in which the eyes EY are aligned for the wearer US; and the +Z direction corresponds to a forward direction or a front direction for the wearer US. The $\pm Y$ direction is parallel to the vertical axis or the vertical direction. In FIG. **1**, the first region AR1 is a perspective view of the HMD **100** corresponding to the wearing state, and the second region AR2 is a perspective view illustrating the collapsing operation of a first temple **100C**.

(20) The HMD **100** allows the wearer US wearing the same not only to visually recognize the virtual image, but also to observe the outside world image in a see-through manner. The HMD **100** can be communicably coupled to an external device (not illustrated) that is a smartphone or the like via a cable **109**, and can form a virtual image corresponding to the video signal input from such an external device.

(21) The HMD **100** includes, as structural elements, a display device **100A** that forms a virtual image, the first temple **100C** that supports the display device **100A**, and a second temple **100D** that supports the display device **100A**. In this case, the center of the display device **100A** is supported by the nose NZ of the wearer US via a nose pad **61**, with both ends of the display device **100A** supported by the ears ER of the wearer US via the first temple **100C** and the second temple **100D**. As a result, the HMD **100** is stably supported on the head of the wearer US.

(22) The display device **100A** includes a first image-forming device **100a** and a second image-forming device **100b**. The first image-forming device **101a** and the second image-forming device **101b** are portions that form a virtual image for the left eye and a virtual image for the right eye, respectively. The first image-forming device **101a** for the left eye includes a first virtual image-forming optical unit **103a** that covers the front of the wearer US in a see-through manner, and a first image-forming main body portion **105a** that forms image light. The second image-forming device **101b** for the right eye includes a second virtual image-forming optical unit **103b** that covers the front of the wearer US in a see-through manner, and a second image-forming main body portion **105b** that forms image light. The virtual image-forming optical units **103a** and **103b** include a light guide formed of a resin material or the like. In the image-forming main body portions **105a** and **105b**, an optical component or an electronic component is housed in an outer case **105d** formed of a magnesium alloy or the like. Bearing members **21a** are provided in the outer case **105d** as components constituting a hinge **23h** to be described later.

(23) The first temple **100C** on the left side is rotatably supported by a pair of hinges **23h** each including a shaft cylinder member **22a** formed on the root side of the first temple **100C**, a pair of bearing members **21a** fixed to the outer case **105d**, and a core rod (not illustrated) passed through the shaft cylinder member **22a** and the bearing members **21a**. The pair of hinges **23h** are provided spaced apart from each other in the Y direction, which is the vertical direction. The pair of hinges **23h** make the first temple **100C** collapsible to the inner side where the head of the wearer US is to be located. Rotating the first temple **100C** to such a position as to extend substantially parallel to the display device **100A** causes the first temple **100C** to be stored behind the display device **100A**.

(24) The second temple **100D** on the right side is rotatably supported by a pair of hinges **23h** each including a shaft cylinder member **22b** formed on the root side of the second temple **100D**, a pair of bearing members **21b** fixed to the outer case **105d**, and a core rod (not illustrated) passed through the shaft cylinder member **22b** and the bearing members **21b**. The pair of hinges **23h** are provided spaced apart from each other in the Y direction, which is the vertical direction. The pair of hinges **23h** make the second temple **100D** collapsible to the inner side where the head of the wearer US is to be located. Rotating the second temple **100D** to such a position as to extend substantially parallel to the display device **100A** causes the second temple **100D** to be stored behind the display device **100A**.

(25) The first temple **100C** includes a rod-shaped first extending member **31a** having an elongated thin plate portion, and a first support member **32a** having a triangular plate shape. Furthermore, the second temple **100D** includes a rod-shaped second extending member **31b** having an elongated thin

plate portion, and a second support member **32b** having a triangular plate shape. The first support member **32a** is rotatably fixed to the first image-forming main body portion **105a** constituting the display device **100A** via the pair of hinges **23h**. The second support member **32b** is rotatably fixed to the second image-forming main body portion **105b** constituting the display device **100A** via the pair of hinges **23h**. The display device **100A** combined with the first support member **32a** and the second support member **32b** is referred to as a display unit **201**. Furthermore, the first extending member **31a** and the second extending member **31b** attached to the display unit **201** are referred to as a wearing unit **202**.

(26) In the first temple **100C**, when the first temple **100C** is expanded to the wearing state, the first extending member **31a** is capable of rotating, relative to the first support member **32a** or the display unit **201**, in a rotational direction about a rotation axis **RX** that is parallel to the **X** direction in which the eyes **EY** are aligned and that corresponds to the first direction. In the second temple **100D**, when the second temple **100D** is expanded to the wearing state, the second extending member **31b** is capable of rotating, relative to the second support member **32b** or the display unit **201**, in a rotational direction about the rotation axis **RX** that is parallel to the **X** direction in which the eyes **EY** are aligned and that corresponds to the first direction.

(27) FIG. 4 is a partially cutaway perspective view illustrating a cross section parallel to the **XZ** plane of the first temple **100C**. FIG. 5 is an exploded perspective view of the first temple **100C**. The first support member **32a** has a thin extending case shape, is formed of resin or the like, and houses the tip portion **31f** of the first extending member **31a**. The first support member **32a** includes a first member **34** disposed on the outer **+X** side, and a second member **35** disposed on the inner **-X** side. The first extending member **31a** is a rod-shaped member, and is formed of a metal or the like. A pair of bearing members **36** and **37**, a sliding member **41**, and a spring **43** that is an elastic member enclosed thereby, are attached to the tip portion **31f** of the first extending member **31a**. One end of the spring **43** is held by a hole **39h** formed by the bearing members **36** and **37**. The other end of the spring **43** is held by a shaft **41h** extending from the sliding member **41**. The sliding member **41** has a groove **41g** on the upper side and the lower side thereof, respectively. This allows the sliding member **41** to move back and forth in the $\pm Z$ direction within an elongated opening **31o** formed at the tip portion **31f** of the first extending member **31a**. Here, the sliding member **41** is a positioning member, and the spring **43** is an elastic member. In other words, in the first extending member **31a**, the sliding member **41** that is a positioning member and the spring **43** that is an elastic member are provided, in addition to the bearing members **36** and **37**, in an opening portion **211a** including the opening **31o**. The sliding member **41** is biased toward the tip **+Z** side by the spring **43**. However, a stopper **31s** at the tip portion **31f** of the first extending member **31a** restricts the sliding member **41** from moving toward the **+Z** side by not less than a predetermined amount. The sliding member **41** can be retracted to the **-Z** side against the spring force. The rear portion **31v** of the first extending member **31a** has some flexibility, and is covered by a soft cover **31q** made of resin (see FIG. 1).

(28) The first member **34** of the first support member **32a** houses the tip portion **31f** of the first extending member **31a** so as to support the same from the outside, that is, from the **+X** side. The second member **35** houses the tip portion **31f** of the first extending member **31a** so as to hold the same from the inside, that is, from the **-X** side. A shaft portion **34x** extending along the rotation axis **RX** is formed on the inner space side of the first member **34**. At the tip portion **31f** of the first extending member **31a**, the elongated opening **31o** is formed, and the pair of bearing members **36** and **37** are attached so as to sandwich a widened portion **31p** provided on the rear end side, that is, on the **-Z** side of the opening **31o**. At this time, a protrusion **38a** formed on the pair of bearing members **36** and **37** are fitted to the widened portion **31p**, and the pair of bearing members **36** and **37** are fixed to each other by utilizing a pawl **38d** and an opening **38f** formed therein. Thus, the bearing members **36** and **37** are integrally fixed to the widened portion **31p**, that is, the first extending member **31a**. A shaft hole **38x** is formed in the pair of bearing members **36** and **37**. The

shaft portion **34x** of the first support member **32a** is inserted into the shaft hole **38x**. Here, the shaft portion **34x** functions as a first rotation mechanism **203a** that allows rotation of the first extending member **31a** relative to the display unit **201**, that is, rotation of the display unit **201** relative to the first extending member **31a** or the wearing unit **202**. Furthermore, the bearing members **36** and **37** function as a first fitting mechanism **204a** fitted to the shaft portion **34x**, which is the first rotation mechanism **203a**. In other words, the first rotation mechanism **203a** and the first fitting mechanism **204a** make the first extending member **31a** rotatable relative to the first support member **32a** about the rotation axis RX corresponding to the first direction.

(29) A ratchet tooth member **45** is formed on the inner surface on the tip side, that is, on the +Z side of the first member **34** of the first support member **32a** along an arc about the shaft portion **34x**. Five recesses **46d** are formed in the ratchet tooth member **45**. A protrusion **46p** of a ratchet pawl **41a** formed in the sliding member **41** incorporated into the first extending member **31a** engages with a recess **46d**. Applying a rotational force greater than or equal to a predetermined magnitude to the first extending member **31a** to rotate the first extending member **31a** about the rotation axis RX relative to the first support member **32a** causes the protrusion **46p** of the ratchet pawl **41a** to move in a stepwise manner while being fitted to recesses **46d** along the ratchet tooth member **45**. This causes the angular posture of the first extending member **31a** relative to the first support member **32a** to change in units of 4° , for example, with the initial standard state being 0° . Specifically, the angular posture of the first extending member **31a** can be -8° , -4° , 0° , $+4^\circ$, or $+8^\circ$. In the above description, the ratchet pawl **41a** is a first positioning portion **205a**, which is one member for adjusting the angle of the first extending member **31a** relative to the first support member **32a**. The ratchet pawl **41a** is provided with a protrusion **46p**, which is brought into contact with any of a plurality of recesses **46d** of the ratchet tooth member **45**. Furthermore, the ratchet tooth member **45** is a second positioning portion **206a**, which is the other member for adjusting the angle of the first extending member **31a** relative to the first support member **32a**. The protrusion **46p** of the ratchet pawl **41a** is brought into contact with any of the plurality of recesses **46d**. Note that the protrusion **46p** of the ratchet pawl **41a** is separated into two portions in the lateral X direction, with the stopper **31s** of the first extending member **31a** being disposed in a groove formed therebetween.

(30) FIG. 6 is a partially cutaway perspective view illustrating a cross section parallel to the XZ plane of the second temple **100D**. FIG. 7 is an exploded perspective view of the second temple **100D**. The second support member **32b** has a thin extending case shape and houses the tip portion **31f** of the second extending member **31b**. The second support member **32b** includes a first member **34** disposed on the outer -X side, and a second member **35** disposed on the inner +X side. A pair of bearing members **36** and **37**, a sliding member **41**, and a spring **43** enclosed thereby are attached to the tip portion **31f** of the second extending member **31b**. One end of the spring **43** is held by a hole **39h** formed by the bearing members **36** and **37**. The other end of the spring **43** is held by a shaft **41h** extending from the sliding member **41**. The sliding member **41** has a groove **41g** on the upper side and the lower side, respectively. This allows the sliding member **41** to be move back and forth in the $\pm Z$ direction within an elongated opening **310** formed at the tip portion **31f** of the first extending member **31a**. Here, the sliding member **41** is a positioning member, and the spring **43** is an elastic member. In other words, in the second extending member **31b**, the sliding member **41** that is a positioning member and the spring **43** that is an elastic member are provided, in addition to the bearing members **36** and **37**, in an opening portion **211b** including the opening **310**. The sliding member **41** is biased toward the tip +Z side by the spring **43**. However, a stopper **31s** at the tip portion **31f** of the first extending member **31a** restricts the sliding member **41** from moving toward the +Z side by not less than a predetermined amount. The sliding member **41** can be retracted to the -Z side against the spring force.

(31) The first member **34** of the second support member **32b** houses the tip portion **31f** of the second extending member **31b** so as to support the same from the outside, that is, from the -X side.

The second member **35** houses the tip portion **31f** of the second extending member **31b** so as to hold the same from the inside, that is, from the +X side. A shaft portion **34x** extending along the rotation axis RX is formed on the inner space side of the first member **34**. At the tip portion **31f** of the second extending member **31a**, the elongated opening **31o** is formed, and the pair of bearing members **36** and **37** are attached so as to sandwich a widened portion **31p** provided on the rear end side, that is, on the -Z side of the opening **31o**. At this time, a protrusion **38a** formed on the pair of bearing members **36** and **37** are fitted to the widened portion **31p**, and the pair of bearing members **36** and **37** are fixed to each other by utilizing a pawl **38d** and an opening **38f** formed therein. Thus, the bearing members **36** and **37** are integrally fixed to the widened portion **31p**, that is, the second extending member **31b**. A shaft hole **38x** is formed in the pair of bearing members **36** and **37**. The shaft portion **34x** of the second support member **32b** is inserted into the shaft hole **38x**. Here, the shaft portion **34x** functions as a second rotation mechanism **203b** that allows rotation of the second extending member **31b** relative to the display unit **201**, that is, rotation of the display unit **201** relative to the second extending member **31b** or the wearing unit **202**. Furthermore, the bearing members **36** and **37** function as a second fitting mechanism **204b** fitted to the shaft portion **34x**, which is the second rotation mechanism **203b**. In other words, the second rotation mechanism **203b** and the second fitting mechanism **204b** make the second extending member **31b** rotatable relative to the second support member **32b** about the rotation axis RX corresponding to the first direction.

(32) A ratchet tooth member **45** is formed on the inner surface on the tip side, that is, on the +Z side of the first member **34** of the second support member **32b** along an arc about the shaft portion **34x**. Five recesses **46d** are formed in the ratchet tooth member **45**. A protrusion **46p** of a ratchet pawl **41a** formed in the sliding member **41** incorporated into the second extending member **31b** engages with a recess **46d**. Applying a rotational force greater than or equal to a predetermined magnitude to the second extending member **31b** to rotate the second extending member **31b** about the rotation axis RX relative to the second support member **32b** causes the protrusion **46p** of the ratchet pawl **41a** to move in a stepwise manner while being fitted to recesses **46d** along the ratchet tooth member **45**. This causes the angular posture of the second extending member **31b** relative to the second support member **32b** to change in units of 4°, for example, with the initial standard state being 0°. Specifically, the angular posture of the second extending member **31b** can be -8°, -4°, 0°, +4°, or +8°. In the above description, the ratchet pawl **41a** is a third positioning portion **205b**, which is one member for adjusting the angle of the second extending member **31b** relative to the second support member **32b**. The ratchet pawl **41a** is provided with a protrusion **46p**, which is brought into contact with any of a plurality of recesses **46d** of the ratchet tooth member **45**. Furthermore, the ratchet tooth member **45** is a fourth positioning portion **206b**, which is the other member for adjusting the angle of the second extending member **31b** relative to the second support member **32b**. The protrusion **46p** of the ratchet pawl **41a** is brought into contact with any of the plurality of recesses **46d**.

(33) An internal structure and the like of the first image-forming device **101a** and the second image-forming device **101b** constituting the display device **100A** will be described with reference to FIG. 8. The first image-forming main body portion **105a** for the left eye holds a display element **71**, a projection lens **72**, a first electronic printed wired board **73**, a second electronic printed wired board **74**, and the like in an outer case **105d** that has a cover shape and that is made of a metal. The display element **71** and the projection lens **72** are fixed in the outer case **105d** in a state of being aligned. In particular, the projection lens **72** is fixed in a state of being also aligned with the tip portion of the first virtual image-forming optical unit **103a**. The first electronic printed wired board **73** has a function of interfacing with an external device (not illustrated). The first electronic printed wired board **73** also manages and controls the display operation of the second electronic printed wired board **74**. The second electronic printed wired board **74** is a driving printed wired board that drives the display element **71** in the first image-forming main body portion **105a**. The second electronic printed wired board **74** operates under the control of the first electronic printed wired

board 73.

(34) The second image-forming main body portion **105b** for the right eye holds a display element **71**, a projection lens **72**, a second electronic printed wired board **74**, and the like in an outer case **105d**. The display element **71** and the projection lens **72** are fixed in the outer case **105d** in a state of being aligned. In particular, the projection lens **72** is fixed in a state of being also aligned with the tip portion of the second virtual image-forming optical unit **103b**. The second electronic printed wired board **74** is a driving printed wired board that drives the display element **71** in the second image-forming main body portion **105b**. The second electronic printed wired board **74** operates under the control of the first electronic printed wired board **73** provided in the distant first image-forming main body portion **105a**.

(35) The first virtual image-forming optical unit **103a** and the second virtual image-forming optical unit **103b** are not separate bodies, but are coupled to each other at opposing ends thereof to form a see-through type light-guiding unit **103c** that is an integrated member. The see-through type light-guiding unit **103c** includes a pair of light-guiding members **10a** and **10b** that guide image light from the display element **71**, and a central member **3** that allows a superimposed view of the outside world image. The pair of light-guiding members **10a** and **10b** are a pair of optical members that contribute to virtual image formation while allowing image light to propagate therein. The central member **3** includes a pair of light-transmitting portions **3a** and **3b**. One light-transmitting portion **3a** is joined to one light-guiding member **10a**. The other light-transmitting portion **3b** is joined to the other light-guiding member **10b**. The see-through type light-guiding unit **103c** is a composite light-guiding device that provides images for both eyes to the wearer US by guiding light. Both ends, that is, the tips of the light-guiding members **10a** and **10b** are fitted into the outer cases **105d**, and are supported by the image-forming main body portions **105a** and **105b**.

(36) An upper cover **4** is fixed to the upper surface of the see-through type light-guiding unit **103c** so as to cover the same. A gap-like space is formed between the upper cover **4** and the see-through type light-guiding unit **103c**, in which a signal line **5** electrically coupling the first image-forming main body portion **105a** and the second image-forming main body portion **105b** extends.

(37) The display element **71** incorporated into the first image-forming main body portion **105a** is a self-emitting display device that allows for two-dimensional display, and operates in a dot matrix manner. Specifically, organic electroluminescence (EL) display panels are envisaged for the display elements **71**. However, the display elements **71** are by no means limited thereto, and may be panels for liquid crystal display (LCD). When a panel for LCD is used, a matching illumination source is required. The display elements **71** are driven by the first electronic printed wired board **73** to form a color image on a rectangular display surface. The display elements **71** can display a two-dimensional moving image or still image.

(38) FIG. **9** is a view illustrating a portion of the first image-forming device **101a**. In particular, FIG. **9** illustrates the optical structure of the first virtual image-forming optical unit **103a**. As described previously, the wearable image display device (HMD) **100** is constituted by the first image-forming device **101a** and the second image-forming device **101b** (see FIG. **1** and the like). However, the first image-forming device **101a** and the second image-forming device **101b** are laterally symmetric and have equivalent structures. Thus, only the first image-forming device **101a** will be described, with description of the second image-forming device **101b** being omitted. Note that in FIG. **9**, x, y, and z represent an orthogonal coordinate system: the x direction and the y direction are parallel to a first surface **S11** and a third surface **S13**; and the z direction is perpendicular to the first surface **S11** and the third surface **S13**.

(39) Of the first virtual image-forming optical unit **103a**, the light-guiding member **10a** is joined to the light-transmitting portion **3a** via an adhesive layer CC. In other words, a second transmission surface **S52** of the light-transmitting portion **3a** is disposed opposed to a second surface **S12** of the light-guiding member **10a**, and has the same shape. The light-guiding member **10a** and the light-transmitting portion **3a** have a structure in which a surface of a main body member that provides a

three-dimensional shape including an optical surface is coated with a thin hard coat layer. The main body members of the light-guiding member **10a** and the light-transmitting member **3a** are formed of a resin material that exhibits high optical transparency in a visible region.

(40) Hereinafter, an overview of the optical path of image light ML will be described. The light-guiding member **10a** guides the image light ML emitted from the projection lens **21** toward an eye of the wearer US by reflection or the like at the first to fifth surfaces **S11** to **S15**. Specifically, the image light ML from the projection lens **21** is first incident on a portion of a fourth surface **S14** formed in a light-incident portion **11a** and is reflected by the fifth surface **S15** that is an inner surface of a reflection film RM, is incident again on the fourth surface **S14** from the inside and is totally reflected, is incident on the third surface **S13** and is totally reflected, and is incident on the first surface **S11** and is totally reflected. The image light ML totally reflected by the first surface **S11** is incident on the second surface **S12**, is partially reflected while partially passing through a half mirror **15** provided on the second surface **S12**, and is incident again on a portion of the first surface **S11** formed in a light-emitting portion **11b** to pass therethrough. The image light ML that has passed through the first surface **S11** travels as a whole along an optical axis AX that is substantially parallel to the Z direction, and is incident on the exit pupil EP at which the eye of the wearer US is disposed as a substantially parallel light flux. In other words, the wearer US observes the image formed by the image light as the virtual image.

(41) The first virtual image-forming optical unit **103a** is configured to allow the wearer US to visually recognize the image light ML by the light-guiding member **10a**, and to allow the wearer US to observe the outside world image with little distortion with the light-guiding member **10a** and the light-transmitting member **3a** being combined. At this time, the third surface **S13** and the first surface **S11** are flat surfaces substantially parallel to each other. Thus, the diopter is substantially 0 for observation performed by seeing through this portion, and aberration or the like is hardly generated for outer light OL. Furthermore, a third transmission surface **S53** and a first transmission surface **S51** are flat surfaces substantially parallel to each other. Moreover, because the third transmission surface **S23** and the first surface **S11** are flat surfaces substantially parallel to each other, aberration or the like is hardly generated. In this way, the wearer US observes the outside world image without distortion through the light-transmitting member **3a**.

(42) FIG. **10** is a view illustrating angle adjustment or position adjustment of the first temple **100C** when wearing the HMD **100**. In the first temple **100C**, the angular posture of the first extending member **31a** relative to the first support member **32a** can be set, with the state indicated by a solid line as the standard state **S0**, to an upper end state **S1** in which a rear end portion **31r** of the first extending member **31a** is displaced counterclockwise upward as indicated by a dot-dash line, and to a lower end state **S2** in which the rear end portion **31r** of the first extending member **31a** is displaced clockwise downward as indicated by a two-dot-dash line. For the first extending member **31a**, the standard state **S0** can be considered to correspond to the first position in the rotational direction, and the upper end state **S1** or the lower end state **S2** can be considered to correspond to the second position in the rotational direction. However, this does not necessarily apply. The lower end state **S2** may be considered to correspond to the first position in the rotational direction and the upper end state **S1** may be considered to correspond to the second position in the rotational direction, and vice versa. As described previously, the angle of the first extending member **31a** can be changed in units of 4° with a click feeling. A reference line **A1** of the upper end state **S1** is raised by two steps from a reference line **A0** of the standard state **S0**, and thus forms an angle $\theta = -8^\circ$ therewith. A reference line **A2** of the lower end state **S2** is lowered by two steps from the reference line **A0** of the standard state **S0**, and thus forms an angle $\theta = +8^\circ$ therewith. Here, the angle θ in the clockwise direction when viewed from the outside (+X side) is positive. In this case, the range of the center angle (absolute value) formed between the first position corresponding to the lower end state **S2** and the second position, which is the standard state **S0**, the upper end state **S1**, or the like, is not less than 4° and not greater than 16° . When the angle θ is approximately $\pm 4^\circ$, the

raised/lowered amount of the first extending member **31a** near the ear ER is approximately ± 5 mm. When the angle θ is approximately $\pm 8^\circ$, the raised/lowered amount of the first extending member **31a** near the ear ER is approximately ± 10 mm. As described above, changing the angular posture of the first extending member **31a** in a stepwise manner can displace the rear end portion **31r** of the first extending member **31a** in the vertical direction parallel to the $\pm Y$ direction, and adapt the HMD **100** to various wearers US of whom the height of the ears ER relative to the eyes EY varies.

(43) FIG. **11** is a view illustrating angle adjustment or position adjustment of the second temple **100D**. Similar to the first temple **100C**, the second temple **100D** can also be adjusted in angle. Specifically, in the second temple **100D**, the angular posture of the second extending member **31b** relative to the second support member **32b** can be set, with the state indicated by a solid line as the standard state **S0**, to an upper end state **S1** in which a rear end portion **31r** of the second extending member **31b** is displaced clockwise upward as indicated by a dot-dash line, and to a lower end state **S2** in which the rear end portion **31r** of the second extending member **31b** is displaced counterclockwise downward as indicated by a two-dot-dash line. For the second extending member **31b**, the standard state **S0** can be considered to correspond to the third position in the rotational direction, and the upper end state **S1** or the lower end state **S2** can be considered to correspond to the fourth position in the rotational direction. However, this does not necessarily apply. The lower end state **S2** may be considered to correspond to the third position in the rotational direction and the upper end state **S1** may be considered to correspond to the fourth position in the rotational direction, and vice versa. As described previously, the angle of the second extending member **31b** can be changed in units of 4° with a click feeling. A reference line **A1** of the upper end state **S1** is raised by two steps from a reference line **A0** of the standard state **S0**, and thus forms an angle $\theta = +8^\circ$ therewith. A reference line **A2** of the lower end state **S2** is lowered by two steps from the reference line **A0** of the standard state **S0**, and thus forms an angle $\theta = -8^\circ$ therewith. Here, the angle θ in the clockwise direction is positive when viewed from the outside ($-X$ side). In this case, the range of the center angle (absolute value) formed between the third position corresponding to the lower end state **S2** and the fourth position, which is the standard state **S0**, the upper end state **S1**, or the like, is not less than 4° and not greater than 16° . As described above, changing the angular posture of the second extending member **31b** in a stepwise manner can displace the rear end portion **31r** of the second extending member **31b** in the vertical direction parallel to the $\pm Y$ direction, and adapt the HMD **100** to various wearers US of whom the height of the ears ER relative to the eyes EY varies.

(44) Note that the angle state of the second temple **100D** can be different from the angle state of the first temple **100C**. This makes it possible to increase the adaptability of the HMD **100** to those wearers US whose left and right ears ER differ in height. Here, the range of the center angle (absolute value) formed between the first position corresponding to the lower end state **S2** of the first support member **32a** and the third position, which is the standard state **S0** of the second support member **32b** or the like, can be not less than 0° and not greater than 8° , for example. In this case, when the angle state of the second temple **100D** and that of the first temple **100C** are varied, the range of the center angle (absolute value) formed between the first position corresponding to the lower end state **S2** of the first support member **32a** and the fourth position, which is the upper end state **S1** of the second support member **32b** or the like, is not less than 4° and not greater than 16° . When such is the range, the posture of the display device **100A** relative to the first temple **100C** and the second temple **100D** can be adjusted while the angle state of the second temple **100D** is made different from the angle state of the first temple **100C**. Posture adjustment of the display device **100A** will be described in detail later.

(45) In the above description, the nose pad **61** can be replaced or deformed, so the display device **100A** can be properly disposed in front of the eyes EY without obstructing observation of the virtual image.

(46) The rotation axis RX of the first extending member **31a** and the second extending member **31b**

may be disposed rearward of (on the $-Z$ side of) the front portion of the eyes EY corresponding to the exit pupil EP, and may be disposed in the rear of the eyes EY. Disposing the rotation axis RX relatively rearward in this manner sufficiently separates the rotation axis RX from the first virtual image-forming optical unit **103a** and the like, allowing the stepwise upward/downward movement of the display image to be made gradual and moderate. Making the stepwise upward/downward movement of the display image gradual can suppress the occurrence of a phenomenon in which the display image becomes invisible or missing.

(47) FIG. **12** is a view illustrating posture adjustment of the display device **100A** relative to the first temple **100C** and the like. In FIG. **12**, the first region BR1 illustrates a standard posture of the display device **100A**; the second region BR2 illustrates the maximum downward posture of the display device **100A**; and the third region BR3 illustrates the maximum upward posture of the display device **100A**. In the first region BR1, the optical axis AX0 of the display device **100A** is parallel to the reference axis TX of the first extending member **31a** of the first temple **100C**, and coincides with the standard line-of-sight direction of the wearer US. In the second region BR2, the optical axis AX0 of the display device **100A** forms an angle $\theta=+8^\circ$ with the reference axis TX of the first extending member **31a** of the first temple **100C**, and is considerably more downward than the standard line-of-sight direction of the wearer US. In the third region BR3, the optical axis AX0 of the display device **100A** forms an angle $\theta=-8^\circ$ with the reference axis TX of the first extending member **31a** of the first temple **100C**, and is considerably more upward than the standard line-of-sight direction of the wearer US. Note that although illustration is omitted, the angle θ can be adjusted in units of 4° within the range of -8° to $+8^\circ$. As described above, the optical axis AX0 of the display device **100A** can be changed in a stepwise manner relative to the reference axis TX of the first extending member **31a**. Thus, the direction of the virtual image visible to the wearer US can be moved upward and downward in a stepwise manner, making it possible to adjust height so as to display the image at a height position preferred by the wearer US. Although the above description has not touched on the second temple **100D**, the second temple **100D** is set to the same state as that of the first temple **100C**. However, this does not apply when the left and right ears ER of the wearer US differ in height. For example, when the difference in height between the left and right ears ER corresponds to an angular difference of 8° between the first extending member **31a** and the second extending member **31b**, the posture adjustment angle of the display device **100A** relative to the first temple **100C** and the like as described above is 8° .

(48) In the wearable image display device or the HMD **100** according to the first embodiment described above, the first positioning portion **205a** and the second positioning portion **206a** are brought into contact with each other at the first position and the second position in the rotational direction that are different from each other. Here, the first positioning portion **205a** and the second positioning portion **206a** are brought into contact with each other either at the first position or at the second position in terms of timing, but are structurally brought into contact with each other both at the first position and at the second position. In other words, in the HMD **100** according to the embodiment, the wearing unit **202** can be set to a plurality of angle states relative to the display unit **201** including a first angle state corresponding to the first position (e.g., the standard state S0 in FIG. **10**) and a second angle state corresponding to the second position (e.g., the upper end state S1 in FIGS. **10** and **11**), with the angle state of the display unit **201** relative to the wearing unit **20**, that is, the disposition of the display unit **201** relative to the ears ER or the eyes EY of the wearer US fixed in a stable state.

Second Embodiment

(49) Hereinafter, a second embodiment of a wearable image display device according to the present disclosure will be described with reference to drawings. The wearable image display device according to the second embodiment is a partial modification to the wearable image display device according to the first embodiment, so description of common parts is omitted.

(50) FIG. **13** is a view illustrating the appearance of an HMD **100** that is the wearable image

display device according to the second embodiment. In FIG. 13, the first region CR1 is a plan view of the HMD 100; the second region CR2 is a front view of the HMD 100; and the third region CR3 is a side view of the HMD 100.

(51) The HMD 100 includes a display device 100A disposed so as to cover the front of the wearer US, and a support device 100S that supports the display device 100A. Functionally speaking, the display device 100A includes a first image-forming device 101a and a second image-forming device 101b. The first image-forming device 100a includes a first display driving unit 102a disposed at an upper portion thereof, and a first combiner 102i that has a spectacle lens shape and that covers the front of the eye. Similarly, the second image-forming device 100b is constituted by a second display driving unit 102b disposed at an upper portion thereof, and a second combiner 102j that has a spectacle lens shape and that covers the front of the eye. The support device 100S includes a main body frame 100F, a first temple 100C, and a second temple 100D. In the above description, the display device 100A is also referred to as the display unit 201, while the support device 100S is also referred to as the wearing unit 202. In the support device 100S, the first temple 100C functions as a first extending member 202a, while the second temple 100D functions as a second extending member 202b.

(52) The first image-forming device 101a and the second image-forming device 101b that constitute the display device 100A are coupled to each other and integrated, and are covered from above by a dome-shaped upper outer packaging member 81 that is elongated in the lateral direction, and from below by a flat plate-shaped lower outer packaging member 82. The first combiner 102i and the second combiner 102j have a hemisphere shape with the upper portion thereof cut off, are supported by the lower outer packaging member 82, and are disposed so as to protrude downward from the lower outer packaging member 82, respectively.

(53) FIG. 14 is a partially enlarged perspective view of the HMD 100. In the figure, the first region DR1 illustrates a coupling portion between the display device 100A and the main body frame 100F and the periphery thereof; and the second region DR2 illustrates a coupling portion between the display device 100A and the main body frame 100F and the periphery thereof.

(54) In both ends of the upper outer packaging member 81 of the display device 100A, a pair of shaft portions 81x for support purposes are formed. These shaft portions 81x are rotatably supported by a pair of bearings 85x formed by members 85a fixed to the main body frame 100F of the support device 100S (see FIG. 13). The pair of shaft portions 81x extend in parallel in the X direction. The display device 100A rotates about a rotation axis RX parallel to the X direction, with the shaft portions 81x or the bearings 85x as the center. Furthermore, the first temple 100C and the second temple 100D are collapsibly coupled to the main body frame 100F via a pair of hinges 87h formed by the members 85a fixed to the main body frame 100F. Here, the shaft portion 81x on the first temple 100C side functions as a first rotation mechanism 203a that allows rotation of the first extending member 202a relative to the display unit 201, that is, rotation of the display unit 201 relative to the first extending member 202a or the wearing unit 202. The bearing 85x on the first temple 100C side functions as a first fitting mechanism 204a fitted to the shaft portion 81x that is the first rotation mechanism 203a. On the other hand, the shaft portion 81x on the second temple 100D side functions as a second rotation mechanism 203b that allows rotation of the second extending member 202b relative to the display unit 201, that is, rotation of the display unit 201 relative to the second extending member 202b or the wearing unit 202. The bearing 85x on the second temple 100D side functions as a second fitting mechanism 204b fitted to the shaft portion 81x that is the second rotation mechanism 203b.

(55) FIG. 15 is a view illustrating a cross section substantially parallel to the XZ plane of the HMD 100. In the display device 100A, one shaft portion 81x is hollow. Wiring 6 extending from an electronic printed wired board 373 housed in the display device 100A passes inside the shaft portion 81x supported by the bearing 85x, extending to the first temple 100C and being drawn out to the outside as a cable 109.

(56) FIG. 16 is a partially cutaway cross-sectional view illustrating an angle adjustment mechanism incorporated between the display unit **201** and the wearing unit **202**, in which the cutaway cross section is marked by a diagonal hatching pattern. At the center in the lateral direction ($\pm X$ direction) of the upper outer packaging member **81**, in a front portion **81cf** on the $+Z$ side, there is formed a ratchet tooth member **45** extending substantially parallel to the XY plane. In a depression on the $-Z$ side, that is, on the back surface side of the central portion **88a** of the main body frame **100F**, there is formed a support frame **88d** extending in the $-Z$ direction. This support frame **88d** holds a sliding member **41**, with a spring **43** sandwiched between the bottom of the support frame **88d** and the sliding member **41**. In the ratchet tooth member **45**, there are formed three recesses **46d** aligned in the vertical direction. The protrusion **46p** of the ratchet pawl **41a** formed in the sliding member **41** of the main body frame **100F** engages with any of the recesses **46d**. Applying a force greater than or equal to a predetermined magnitude for rotating the wearing unit **202** about the rotation axis RX relative to the display unit **201** causes the protrusion **46p** of the ratchet pawl **41a** to move upward or downward in a stepwise manner to any of the recesses **46d** while being fitted to recesses **46d** along the ratchet tooth member **45**. This causes the angular posture of the display unit **201** relative to the first extending member **202a** or the second extending member **202b**, that is, the angular posture of the display unit **201** relative to the wearing unit **202** to change in a stepwise manner, for example, in predetermined angle units of 4° . In the above description, the ratchet pawl **41a** is a first positioning portion **205a**, which is one member for adjusting the angle of the wearing unit **202** relative to the display unit **201**. The ratchet pawl **41a** is provided with a protrusion **46p**, which is brought into contact with any of a plurality of recesses **46d** of the ratchet tooth member **45**. Furthermore, the ratchet tooth member **45** is a second positioning portion **206a**, which is the other member for adjusting the angle of the wearing unit **202** relative to the display unit **201**. The protrusion **46p** of the ratchet pawl **41a** is brought into contact with any of the plurality of recesses **46d**.

(57) FIG. 17 is a side cross-sectional view illustrating an optical structure of the first image-forming device **101a**. The first image-forming device **101a** includes a display element **71** and an imaging optical system **172**. The imaging optical system **172** includes a projection lens **172a**, a prism mirror **172b**, a wedge-shaped optical element **172c**, and a see-through mirror **172d**. Of the imaging optical system **172**, the see-through mirror **172d** corresponds to the first combiner **102i**. Note that the optical structure of the second image-forming device **101b** is the same as the optical structure of the first image-forming device **101a**, so description thereof is omitted.

(58) To describe the optical path, the image light ML from the display element **71** is incident on the projection lens **172a**, and is emitted from the projection lens **172a** in a substantially collimated state. The image light ML that has passed through the projection lens **172a** is incident on the prism mirror **172b**, passes through an incident surface **2a** while being refracted thereby, is reflected by an inner reflection surface **2b** with a high reflectance close to 100%, and is refracted again by a light emission surface **2c**. The image light ML from the prism mirror **172b** is incident on the see-through mirror **172d** via the wedge-shaped optical element **172c**, and is reflected by an inner reflection surface **2r** with a predetermined reflectance. The image light ML reflected by the see-through mirror **172d** is incident on the exit pupil EP at which the eye EY of the wearer US is disposed. The outer light OL that has passed through the see-through mirror **172d** is also incident on the exit pupil EP.

(59) In the wearable image display device or the HMD **100** according to the second embodiment described above, a single second positioning portion **206a** is disposed away from the first rotation mechanism **203a** and the second rotation mechanism **203b**, which makes it easy to adjust the angle of the display unit **201** relative to the wearing unit **202**.

Modified Examples and Others

(60) The present disclosure has been described based on the embodiments described above. However, the present disclosure is by no means limited to the embodiments described above. The

present disclosure can be carried out in various modes without departing from the gist of the present disclosure. Thus, for example, the following modifications can also be carried out.

(61) In the embodiments described above, the ratchet pawl **41a** is formed in the wearing unit **202**, and the ratchet tooth member **45** is formed in the display unit **201**. However, the ratchet tooth member **45** may be formed in the wearing unit **202**, and the ratchet pawl **41a** may be formed in the display unit **201**. In other words, the protrusion **46p** and the recesses **46d** can be interchangeably disposed.

(62) In the embodiments described above, the shaft portions **81x** can be provided in the display unit **201**, and the bearings **85x** can be provided in the wearing unit **202**. The bearing **85x** and the like are not limited to those illustrated, and can have various structures.

(63) The pitch units of the angle θ of the first extending member **31a** and the second extending member **31b** are not limited to 4° , and can vary depending on the application. The upper limit value is not limited to $\pm 8^\circ$ either, and can vary depending on the application.

(64) The first image-forming device **101a** and the second image-forming device **101b** are not limited to the illustrated optical systems, and can be optical systems including various optical elements.

(65) A dimming device for performing dimming by restricting incident light to the virtual image-forming optical units **103a** and **103b** can be attached on the outside world side of the virtual image-forming optical units **103a** and **103b** and the combiners **102i** and **102j**. The dimming device adjusts transmittance, for example, electrically. Mirror liquid crystals, electronic shades, and the like can be used as the dimming device. The dimming device may be those that adjust transmittance in accordance with outer light brightness.

(66) The combiners **102i** and **102j** can also be replaced with mirrors having light-blocking properties. In this case, a non-see-through type optical system is obtained that does not assume direct observation of the outside world image.

(67) A wearable image display device according to a specific aspect includes: a display unit configured to form an image; and a wearing unit attached to the display unit; wherein the display unit includes a first positioning portion and a first rotation mechanism configured to rotate relative to the wearing unit in a rotational direction with a first direction serving as a rotation axis, the wearing unit includes a second positioning portion brought into contact with the first positioning portion and a first fitting mechanism fitted to the first rotation mechanism, and the first positioning portion and the second positioning portion are brought into contact with each other either at a first position or at a second position different from the first position in the rotational direction.

(68) In the wearable image display device described above, the first positioning portion and the second positioning portion are brought into contact with each other at a first position and a second position in the rotational direction. Thus, the wearing unit can be set to a plurality of angle states relative to the display unit including a first angle state corresponding to the first position and a second angle state corresponding to the second position, with the angle state of the display unit relative to the wearing unit, that is, the disposition of the display unit relative to the ears or eyes of the wearer fixed in a stable state.

(69) In a specific aspect, the first position is a position on a first rotational direction side in the rotational direction, the second position is a position on a second rotational direction side in a direction opposite to the first rotational direction, the first positioning portion is provided with one of a recess or a protrusion, the second positioning portion is provided with the other of the recess or the protrusion, the one provided in the first positioning portion is brought into contact, at the first position, with the other provided in the second positioning portion when the display unit is rotated to the first rotational direction side, and the one provided in the first positioning portion is brought into contact, at the second position, with the other provided in the second positioning portion when the display unit is rotated to the second rotational direction side. In this case, the first positioning portion and the second positioning portion can be easily and reliably positioned by the fitting of a

recess and a protrusion.

(70) In a specific aspect, the first position is a position on a first rotational direction side in the rotational direction, the second position is a position different from the first position in the first rotational direction, the first positioning portion is provided with one of a recess or a protrusion, and the second positioning portion is provided with the other of the recess or the protrusion.

(71) In a specific aspect, a range of a center angle formed between the first position and the second position about the rotation axis in the first direction is not less than 4° and not greater than 16° . In this case, in response to a change from the first position to the second position, the angular posture of the display unit relative to the wearing unit can be changed in a range of not less than 4° and not greater than 16° .

(72) In a specific aspect, the wearing unit includes a first extending member extending in a second direction different from the first direction, the display unit includes a first support member to which the first extending member is attached, the first support member is provided with the first rotation mechanism and the first positioning portion, and the first extending member is provided with the first fitting mechanism and the second positioning portion. In this case, the angle state of the display unit relative to the wearing unit can be adjusted while positioning and fitting the first extending member and the first support member to each other.

(73) In a specific aspect, the first rotation mechanism is a shaft portion with the first direction serving as an axial direction, the first fitting mechanism is a bearing portion fitted to the shaft portion, the first extending member is provided with an opening portion opening in the second direction, and in the first extending member, a bearing member including the bearing portion, a positioning member including the second positioning portion, and an elastic member provided between the bearing member and the positioning member are provided in the opening portion. In this case, the second positioning member can be elastically displaced relative to the bearing member by the elastic member, and the second positioning portion can be biased and fixed to the first positioning portion with an appropriate force. With such a structure, the structures of the first extending member and the first support member can be made thin and small.

(74) In a specific aspect, the wearing unit includes a second extending member that extends in the second direction and that is different from the first extending member, the display unit includes a third positioning member and a second support member to which the second extending member is attached, the first support member is provided on one side in the first direction of the display unit, the second support member is provided on the other side in the first direction of the display unit, the second support member includes a second rotation mechanism configured to rotate relative to the wearing unit in the rotational direction with the first direction serving as the rotation axis, the second extending member includes a second fitting mechanism fitted to the second rotation mechanism and a fourth positioning portion brought into contact with the third positioning portion in the rotational direction, and the third positioning portion and the fourth positioning portion are brought into contact with each other either at a third position or at a fourth position different from the third position in the rotational direction. In this case, the first extending member of the wearing unit is supported via the first support member of the display unit, and the second extending member of the wearing unit is supported via the second support member of the display unit. Thus, the first support member of the wearing unit can be set to a plurality of angle states relative to the display unit, and the second support member of the wearing unit can be set to a plurality of angle states relative to the display unit.

(75) In a specific aspect, the third positioning portion is provided with one of a recess or a protrusion, the fourth positioning portion is provided with the other of the recess or the protrusion, the one provided in the third positioning portion is brought into contact, at the third position, with the other provided in the fourth positioning portion when the display unit is rotated to the first rotational direction side, the one provided in the third positioning portion is brought into contact with the other provided in the fourth positioning portion at the fourth position when the display unit

is rotated to the second rotational direction side, a range of a center angle formed between the first position and the third position about the rotation axis in the first direction is not less than 0° and not greater than 8° , and a range of a center angle formed between the first position and the fourth position about the rotation axis in the first direction is not less than 0° and not greater than 16° . In this case, the relative angle between the first support member of the wearing unit and the second support member of the wearing unit can be in a range of not less than 0° and not greater than 8° at the minimum, and can be in a range of not less than 0° and not greater than 16° at the maximum.

(76) In a specific aspect, the display unit includes a second rotation mechanism configured to rotate relative to the wearing unit in the rotational direction, the first rotation mechanism is provided on one side, in the first direction, of the display unit, the second rotation mechanism is provided on the other side, in the first direction, of the display unit, the wearing unit includes a second fitting mechanism fitted to the second rotation mechanism, a first extending member extending in a second direction different from the first direction, and a second extending member extending in the second direction, the first fitting mechanism and the first extending member are each provided on one side, in the first direction, of the wearing unit, the second fitting mechanism and the second extending member are each provided on the other side, in the first direction, of the wearing unit, and the second positioning portion is provided between the first fitting mechanism and the second fitting mechanism of the wearing unit. In this case, a single second positioning portion is disposed away from the first rotation mechanism and the second rotation mechanism.

Claims

1. A wearable image display device comprising: a display unit that displays an image; and a wearing unit attached to the display unit; wherein the display unit includes a first positioning portion and a first rotation mechanism that rotates relative to the wearing unit in a rotational direction with a first direction serving as a rotation axis, the wearing unit includes a second positioning portion brought into contact with the first positioning portion and a first fitting mechanism fitted to the first rotation mechanism, the first positioning portion and the second positioning portion are brought into contact with each other either at a first position or at a second position different from the first position in the rotational direction, and a range of a center angle formed between the first position and the second position about the rotation axis in the first direction is not less than 4° and not greater than 16° .
2. The wearable image display device according to claim 1, wherein the first position is a position on a first rotational direction side in the rotational direction, the second position is a position on a second rotational direction side in a direction opposite to the first rotational direction, the first positioning portion is provided with one of a recess or a protrusion, the second positioning portion is provided with the other of the recess or the protrusion, the one of the recess or the protrusion provided in the first positioning portion is brought into contact, at the first position, with the other of the recess or the protrusion provided in the second positioning portion when the display unit is rotated to the first rotational direction side, and the one of the recess or the protrusion provided in the first positioning portion is brought into contact, at the second position, with the other of the recess or the protrusion provided in the second positioning portion when the display unit is rotated to the second rotational direction side.
3. The wearable image display device according to claim 1, wherein the first position is a position on a first rotational direction side in the rotational direction, the second position is a position different from the first position in the first rotational direction, the first positioning portion is provided with one of a recess or a protrusion, and the second positioning portion is provided with the other of the recess or the protrusion.
4. The wearable image display device according to claim 1, wherein the wearing unit includes a first extending member extending in a second direction different from the first direction, the

display unit includes a first support member to which the first extending member is attached, the first support member is provided with the first rotation mechanism and the first positioning portion, and the first extending member is provided with the first fitting mechanism and the second positioning portion.

5. The wearable image display device according to claim 4, wherein the first rotation mechanism is a shaft portion with the first direction serving as an axial direction, the first fitting mechanism is a bearing portion fitted to the shaft portion, the first extending member is provided with an opening portion opening in the second direction, and a bearing member including the bearing portion, a positioning member including the second positioning portion, and an elastic member provided between the bearing member and the positioning member are provided in the opening portion of the first extending member.

6. The wearable image display device according to claim 4, wherein the wearing unit includes a second extending member that extends in the second direction and that is different from the first extending member, the display unit includes a third positioning member and a second support member to which the second extending member is attached, the first support member is provided on one side, in the first direction, of the display unit, the second support member is provided on the other side, in the first direction, of the display unit, the second support member includes a second rotation mechanism that rotates relative to the wearing unit in the rotational direction with the first direction serving as the rotation axis, the second extending member includes a second fitting mechanism fitted to the second rotation mechanism and a fourth positioning portion brought into contact with the third positioning portion in the rotational direction, and the third positioning portion and the fourth positioning portion are brought into contact with each other either at a third position or at a fourth position different from the third position in the rotational direction.

7. The wearable image display device according to claim 6, wherein the third positioning portion is provided with one of a recess or a protrusion, the fourth positioning portion is provided with the other of the recess or the protrusion, the one of the recess or the protrusion provided in the third positioning portion is brought into contact, at the third position, with the other of the recess or the protrusion provided in the fourth positioning portion when the display unit is rotated to the first rotational direction side, the one of the recess or the protrusion provided in the third positioning portion is brought into contact, at the fourth position, with the other of the recess or the protrusion provided in the fourth positioning portion when the display unit is rotated to the second rotational direction side, a range of a center angle formed between the first position and the third position about the rotation axis in the first direction is not less than 0° and not greater than 8° , and a range of a center angle formed between the first position and the fourth position about the rotation axis in the first direction is not less than 0° and not greater than 16° .

8. The wearable image display device according to claim 1, wherein the display unit includes a second rotation mechanism that rotates relative to the wearing unit in the rotational direction, the first rotation mechanism is provided on one side, in the first direction, of the display unit, the second rotation mechanism is provided on the other side, in the first direction, of the display unit, the wearing unit includes a second fitting mechanism fitted to the second rotation mechanism, a first extending member extending in a second direction different from the first direction, and a second extending member extending in the second direction, the first fitting mechanism and the first extending member are each provided on one side, in the first direction, of the wearing unit, the second fitting mechanism and the second extending member are each provided on the other side, in the first direction, of the wearing unit, and the second positioning portion is provided between the first fitting mechanism and the second fitting mechanism of the wearing unit.
